Narrated Accessibility Walk

Harvard station to Spaulding Rehabilitation Hospital Cambridge (1575 Cambridge St), Cambridge, MA

Pilot report 4 | Recorded October 27, 2026

Part of an independent series of narrated walks that check whether people with disabilities can reach clinical research sites. Prepared by Uri Kartoun, PhD, Founder, DBbun LLC, Cambridge, MA | uri.kartoun@dbbun.com | dbbun.com

Contents

Summary....1
Walk details....2
What travelers would meet....2
Findings....3
A. Harvard station to the Spaulding gate....3
B. Spaulding entrance and grounds....4
Evidence for the priority findings....5
Questions for the hospital....6
Prospective test: the open-data screen vs. the walk....6
Checking the AI-assisted analysis....7
Photo evidence....7
How this report was produced....10

Summary

A narrated walk documented the trip a research participant might make on foot from the Red Line at Harvard station to Spaulding Rehabilitation Hospital Cambridge, through Harvard Square, Harvard Yard and along Cambridge St, and then around the hospital's entrance and grounds. The route was first screened using only public data, before the walk, so the walk also tests how well the screen predicts what a traveler meets.

- **25 items recorded:** 16 possible barriers, 2 rated high severity, and 9 positive features.
- **Few obstacles were seen at the hospital entrance.** Automatic doors, a bench at the entrance, clear signs, accessible parking at two entrances and two bays for ambulances and wheelchair vans. The walker's verdict on arrival: "highly accessible from the Harvard Square T."
- **The burden is the approach:** a broken tactile warning strip at Harvard Square, brick sections and a sidewalk pothole on Cambridge St, about 4 minutes of walking with no bench, and tree roots on the site path to the outpatient center.
- **Prospective test:** the open-data screen predicted the brick, the seatless stretch and the gentle slope, but over-predicted curb ramp problems from 10 to 20-year-old ratings and missed the day-of defects. Screening and walking together give a fuller picture than either alone.
- **Scope:** the building interior was not assessed. By protocol, the walker does not enter care facilities without an appointment or an invitation. Interior items are listed as questions for the hospital.

Disclaimer. This report was produced with the help of AI tools (automatic speech transcription and AI-assisted image analysis; see “How this report was produced”) and was reviewed by the author. Errors and omissions are possible. Items are possible barriers observed on one day, not legal compliance determinations or findings of violation. Distances are estimates. Observations reflect conditions on October 27, 2026 and may have changed since. This is an independent project; it is not affiliated with or endorsed by the hospital.

Walk details

Date	October 27, 2026 (Tuesday), morning, sunny and dry
Recordings	Video 1: Harvard station to the Spaulding gate (19 min 33 s). Video 2: the Spaulding frontage and main entrance (first 2 minutes used; the rest covers a separate site). Audio: the Spaulding entrance, grounds, parking and outpatient center (8 min 36 s; audio only, no video near patients).
Route	Harvard station elevator, Harvard Square, Harvard Yard, Quincy St, then Cambridge St eastward past Felton St, Irving St, Trowbridge St and Roberts Rd to Spaulding (about 1.2 km)
Narrator	Uri Kartoun, PhD (DBbun LLC); assessed the route from the point of view of wheelchair, walker and cane users, blind and low-vision people, and people who tire easily. The walker does not use a mobility aid.
Why this route	Spaulding is a rehabilitation hospital that runs clinical research with people with disabilities. Harvard station is the nearest Red Line station, so this walk is a likely last leg for participants arriving by train.
Scope	Outdoors only. The walker did not enter the hospital (no appointment or invitation).

What travelers would meet

What each group of travelers would likely meet, based on what the walker observed. These are not test results: no traveler with these disabilities made the trip, so each cell notes what was seen and, where it matters, what remains uncertain.

Traveler	Harvard station to Spaulding	At Spaulding	Main obstacles
Wheelchair (manual or power)	Seen: a working elevator at Harvard; about 1.2 km with brick sections, a broken warning strip and a sidewalk pothole (S5, S11, S12). Uncertain: the effort over this distance in a manual wheelchair; curb ramp slopes not measured	Seen: automatic doors, van bays and accessible parking; tree roots on the path to the outpatient center (S22). Uncertain: the interior	S5, S11, S22
Walker, rollator or cane	Seen: broken bricks, a pothole and wet leaves (S8, S12, S13). Uncertain: fatigue over the distance	Seen: a bench at the entrance; tree roots on the outpatient path (S22)	S8, S12, S22
Blind or low vision	Seen: audible signals at two crossings; a broken warning strip at Harvard Square (S5). Uncertain: wayfinding through Harvard Yard	Seen: clear visual signs. Uncertain: tactile or audible wayfinding; the interior	S5

Traveler	Harvard station to Spaulding	At Spaulding	Main obstacles
People who tire easily or are sensitive to fumes	Seen: about 19 minutes at the walker's pace, with no bench for about 4 minutes (S14). Uncertain: bus 69 as an alternative (not tested)	Seen: benches and a garden; vehicles idling by the entrance bench (S18)	S14, S18

Findings

Severity notes appear in the barrier column; "positive" marks features that worked well. Conf. (confidence): H = seen and recorded on video, or confirmed by data; M = described in the narration only (the grounds were recorded as audio) or not measured.

A. Harvard station to the Spaulding gate

#	Where	Possible barrier or feature	Observation	Most affected	Conf.	Suggested fix
1	Harvard station platform	Elevator sign visible (positive)	A sign to the elevator is visible from the platform. One exit toward the buses is stairs only, but the elevator route is signed.	-	H	-
2	Harvard station elevator to the street	No arrival light	Glass elevator, clean, no bad smell (positive). The call button gives a steady light, but there is no light or sound when the car arrives; you can see it coming only because the shaft is glass.	Deaf or hard of hearing; low vision	H	Add an arrival light and chime
3	Harvard station, digital elevator-status screen	Misleading status for Davis	The screen (with a press-for-audio button, positive) lists elevator outages elsewhere. Davis is shown as working, but Davis has no elevator at its Holland St entrance, so a traveler planning a trip could be misled.	Wheelchair users planning a trip to Davis	H	Show entrances without elevators, not only outages
4	Harvard Square and Cambridge St at Quincy St	Accessible pedestrian signals (positive)	Push buttons with audible signals at both crossings; the walk phase was announced.	-	H	-
5	Harvard Square crossing	Broken tactile warning strip (High)	The red tactile warning strip at the curb ramp is badly broken, with holes and missing pieces, and the surface around it is cracked.	Blind and low vision (warning lost); wheelchair; cane	H	Replace the warning strip and repair the ramp surface
6	Harvard Square, construction area	Pothole at a crossing edge; barriers	A pothole at the edge of a crossing; construction barriers along a brick sidewalk. Passable at the time.	Wheelchair, walker	H	Patch the pothole; keep a clear signed path past the work
7	Harvard Yard	Flat paths with benches (positive)	Mostly flat brick and paved paths with benches; repaired root damage.	-	H	-

#	Where	Possible barrier or feature	Observation	Most affected	Conf.	Suggested fix
8	Harvard Yard path	Broken bricks around a utility cover	Broken and missing bricks around a utility cover in the path.	Wheelchair casters, cane, walker	H	Reset the bricks flush with the cover
9	Harvard Yard and Quincy St	Shared paths with cyclists	Many cyclists on shared paths; the walker had to stay alert.	Low vision; slow walkers	M	Signs asking cyclists to slow down on shared paths
10	Church and design-school buildings on the route	Push-button doors (positive)	Accessible entrances with push-button openers.	-	H	-
11	Cambridge St sidewalks	Brick sections	About 35 to 40% of the Cambridge St stretch is brick (by frame count), some uneven from tree roots (“the bricks are more difficult”).	Wheelchair, walker; people with pain or fatigue	H	Reset uneven bricks; smooth concrete strip in the main walking line
12	Cambridge St near Felton St	Sidewalk pothole; tree-lifted panel	A pothole in the sidewalk, and a concrete panel lifted by a tree that has been partly ground down.	Wheelchair, walker, cane	H	Patch the pothole; finish grinding or replace the panel
13	Cambridge St	Wet leaves	Thick wet leaves on the sidewalk (seasonal).	Anyone with balance or mobility problems	H	Leaf clearing on main walking routes to hospitals
14	Cambridge St, Quincy St to Trowbridge St	No bench for about 4 minutes of walking	No bench from Quincy St until just past Trowbridge St (“the first time that we see a bench”), about 4 minutes at the walker’s pace and longer for a slow walker. More benches after that, outside shops.	People who need to rest	H	Add at least one bench on this stretch
15	Cambridge St crossings, Felton St to Roberts Rd	Curb ramps in good shape (positive)	The walker described these crossings as “good shape” and “flat, good quality”; a store selling mobility aids is on the route.	-	M	-
16	Spaulding frontage, Cambridge St	Construction at the frontage	Spaulding sign visible from the street (positive). Construction fencing, cones and tape along the frontage; the sidewalk was still open and wide.	Low vision (cones); wheelchair if the work expands	H	Keep the sidewalk clear and the work edge marked

B. Spaulding entrance and grounds

#	Where	Possible barrier or feature	Observation	Most affected	Conf.	Suggested fix
17	Main entrance	Automatic doors and a bench (positive)	Automatic doors; a bench right by the entrance, in use. The walker: “highly accessible”.	-	M	-

#	Where	Possible barrier or feature	Observation	Most affected	Conf.	Suggested fix
18	Entrance drive	Vehicles idling at the door	Ambulances and transport vans (one with a wheelchair symbol) waited at the entrance with engines running over several minutes, next to the entrance bench.	People with asthma or chemical sensitivity; anyone on the entrance bench	M	No-idling signs; ask transport providers to switch off engines
19	Beside the entrance and neighboring buildings	Garden, sensor doors, café (positive)	A walkway and garden with a small fountain; flat approaches with sensor-operated doors and benches at the neighboring buildings on the site; a café.	-	M	-
20	Parking by the main entrance	Accessible spaces: door room on one side only	Four signed accessible spaces. Only one has room to open doors fully on both sides; the other three have room on one side only.	Wheelchair users, as drivers or passengers, and ramp vans	M	Mark which side each access aisle serves; add a both-sides space
21	Parking by the main entrance	Bays for ambulances and wheelchair vans (positive)	Two spaces reserved for ambulances and wheelchair vans: a defined loading point off the street.	-	M	-
22	Site path toward the outpatient center	Tree roots across the path (High)	Tree roots make this stretch “extremely difficult to cross with a wheelchair”; the walker used the site road instead.	Wheelchair, walker, cane	M	Repair the path over the roots, or sign a step-free route
23	Site road and outpatient center	Clear signs; accessible spaces (positive)	Signs to the Main Entrance, Ambulance Entrance and Outpatient Center; “Do not enter, use main entrance in front” at a side door. Four more accessible spaces at the outpatient center, whose entrance was “highly accessible”.	-	M	-
24	Rear service area	Unmarked steep ramp	A steep ramp to a door, apparently for equipment, with no sign saying it is not a public entrance.	Wheelchair users who take it for an entrance	M	Sign it “Service only; accessible entrance at …”
25	Side path at the frontage	Cracks; temporary cones	Cracks in a side path; cones around a renovation area.	Wheelchair, walker	H	Repair cracks when the renovation ends

Evidence for the priority findings

Where each priority finding can be checked in the recordings. The recordings are kept by the author and are not published, because they show faces and license plates.

#	Finding	Where to find it (file, time)
5	Broken tactile warning strip, Harvard Square	Video IMG_6454, 3:12–3:15 (photo figure)
12	Sidewalk pothole near Felton St	Video IMG_6454, 14:09–14:26 (photo figure)

#	Finding	Where to find it (file, time)
14	No bench, Quincy St to Trowbridge St	Video IMG_6454, 11:54–16:09
18	Vehicles idling at the entrance	Audio spaulding_audio_2.m4a, 0:44–4:00 (narration only)
22	Tree roots on the path to the outpatient center	Audio spaulding_audio_2.m4a, 5:33–5:46 (narration only)

Questions for the hospital

The interior was not assessed. These questions cover what a participant with a disability would need to know before a research visit:

- Is the front desk at a height a seated person can use, and is there a lowered section?
- Where is the nearest accessible restroom to the main entrance?
- Are loaner wheelchairs available at the entrance?
- Is there a quiet waiting area for people with sensory sensitivities?
- Are there signs from the lobby to research and clinic areas, including tactile or high-contrast signs?
- Do study teams tell participants which entrance, drop-off point and parking to use, and about the tree roots on the path to the outpatient center?
- Are transport providers asked to switch off engines while waiting at the entrance?

Prospective test: the open-data screen vs. the walk

Before the walk, on October 23, 2026, Spaulding and its neighbor CHA Cambridge Hospital were screened automatically using only public data (OpenStreetMap, USGS elevation, City of Cambridge open data on curb ramps, sidewalks, crashes and service requests, and the MBTA), from Harvard station and from the nearest bus 69 stops. The screen’s own route from Harvard station went through Broadway and Prescott St, while the walker went through Harvard Yard. To compare like with like, the same screen was re-run after the walk, unchanged, on the Cambridge St stretch actually walked (Quincy St to Spaulding, 609 m). The Harvard Yard paths are not in the city’s sidewalk data, so that part is not compared. On the screen’s original route, the figures were 46% brick and 667 m without a mapped seat.

Screen prediction	What the walk found	Result
About 40% of the walked Cambridge St stretch on brick	About 35 to 40% of the same stretch is brick (frame count)	Confirmed
Longest stretch without a mapped seat: 229 m	No bench from Quincy St to just past Trowbridge St, about 360 m by the screen’s own distances	Not confirmed: one mapped seat was not seen (it may be private, gone, or not a bench)
14 of 34 curb ramps rated a problem by the city (surveys from 2013 and 2023): slope at Broadway and Prescott St, Sumner St and Trowbridge St; puddling at Irving St; fair condition at Felton St and Roberts Rd	The walker found these crossings in good shape. It was a dry day, so puddling could not be seen; slope was not measured	Not confirmed on a dry day; slope needs measuring

Screen prediction	What the walk found	Result
Gentle elevation; slope not the main barrier	No hills noted	Confirmed
No on-street accessible parking near Spaulding; hospital parking and drop-off are not in any open dataset, so check on site	Accessible spaces at two Spaulding entrances, plus two van and ambulance bays	Resolved on site
For people who tire easily, bus 69 is the realistic option: stops 23 to 74 m from the entrance, on concrete	Not tested on this walk	Not tested
(Not predictable from data)	Broken warning strip (S5), broken bricks (S8), sidewalk pothole (S12), no elevator arrival light (S2), misleading elevator-status screen (S3), construction (S16), tree roots (S22), idling vehicles (S18)	Missed by the screen

What this shows: the screen got the lasting features of the route right (surface type and slope) and pointed to where to look. Its seat data missed at least one gap, its curb ramp ratings did not match what the walker saw on a dry day, and it cannot see day-of conditions. It also named its own blind spot in advance: hospital parking and drop-off, which had to be checked on site. The method therefore uses both steps: the screen to plan, the walk to confirm and to cover what the data cannot see.

Checking the AI-assisted analysis

The item list was drafted with AI help from the transcribed narration and the face-blurred frames, and reviewed by the walker. Corrections from that review: The second video was first assumed to show Spaulding but mostly covers another hospital site, so only its first 2 minutes are used here. The audio files' names were the phone's automatic location labels: of five files, one belongs to Spaulding, one is from the start of the day at another station, and three are from another site, so those four were moved to other reports. The interior had been expected in the audio; it was not recorded, by protocol. No faces are shown.

Photo evidence

Frames were taken from video 1. Faces were blurred before review. The grounds were recorded as audio only, so there are no photos of them.



Figure 1. Harvard station: the elevator call panel. The call button lights, but there is no arrival light (item 2).



Figure 2. Harvard Square: the tactile warning strip at a curb ramp, badly broken with missing pieces (item 5).



Figure 3. Harvard Yard: broken and missing bricks around a utility cover (item 8).



Figure 4. Cambridge St near Felton St: a pothole in the concrete sidewalk (item 12).



Figure 5. Cambridge St: a brick sidewalk made uneven by tree roots, covered in leaves (items 11 and 13).



Figure 6. Cambridge St just past Trowbridge St: the first bench after about 4 minutes of walking (item 14).



Figure 7. The Spaulding sign at 1575 Cambridge St, with construction cones and tape along the frontage (item 16).

How this report was produced

- **Capture:** two narrated smartphone videos and one audio recording made by the author.
- **Local processing on the author's computer:** audio extracted with ffmpeg (imageio-ffmpeg 7.1); narration transcribed with faster-whisper, model medium.en, on a local GPU; one frame sampled every 3 seconds; faces blurred with OpenCV 4.14 before any image left the computer.
- **AI review:** the blurred frames and the transcripts were reviewed by Claude (Anthropic), model claude-opus-5-5, which drafted the item list and this report. The author reviewed the findings.
- **Open-data screen:** OpenStreetMap, USGS elevation and City of Cambridge open data, processed with Python (OSMnx, GeoPandas) on the author's computer.

Limitations

- One walker on one dry day; distances are estimates and slopes were not measured.
- The grounds were recorded as audio only, so those items rest on the narration (Conf. M).
- The building interior was not assessed.
- Items are possible barriers, not legal compliance determinations.